MATH 1000 **Name (Print):** ____________________

Summer 2018

- This exam contains 11 pages (including the front and back of this cover page) and 17 problems. Check to see if any pages are missing.
- You have 90 minutes to complete this exam.

The following rules apply:

- If a question is followed by $\boxed{\text{work}}$, then work of some form is required. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. A correct answer that has no supporting work when $\boxed{\text{work}}$ is required will not receive full credit.
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 135 points total in this exam. If a question has multiple parts, the points for that question are divided evenly among the parts.
- Do not write in the $\boxed{\checkmark \quad + \quad -}$ box, because this is for grading. The grading scale (except True/False) is: $\checkmark$ = 100%, $+$ = 50%, $-$ = 0%

Page:	3	4	5	6	7	8	9	10	Total
Points:	10	10	10	20	20	20	20	25	135
Score:									

1. (5 points) A normal distribution has mean $\mu = 100$ kg and standard deviation $\sigma = 10$ kg. Find the z-value for $x = 113$ kg.

   $\boxed{\checkmark \quad + \quad -}$

   (a) 1.3
   (b) 2
   (c) 1.7
   (d) 0.6
   (e) None of the above.

2. (5 points) A normal distribution has mean $\mu = 100$ points and third quartile $Q_3 = 106.75$ points. Find the z-value for $x = 115$ points.

   $\boxed{\checkmark \quad + \quad -}$

   (a) 1.5
   (b) 2.3
   (c) 1.865
   (d) 0.82
   (e) None of the above.

3. (5 points) Find the z-value of Q_1 in any normal distribution.

$\boxed{\checkmark \quad + \quad -}$

(a) 0.675
(b) -1
(c) 1
(d) -0.675
(e) None of the above.

4. (5 points) In a normal distribution with mean $\mu = 44$ lb, a weight of $x = 77$ lb has a z-value of 3. Find the standard deviation σ.

$\boxed{\checkmark \quad + \quad -}$

(a) $\sigma = 8$ lb
(b) $\sigma = 12$ lb
(c) $\sigma = 4$ lb
(d) $\sigma = 10$ lb
(e) none of the above

5. (5 points) The height of an 18-year old student at your university has an approximate normal distribution with a mean of 68 inches and a standard deviation of 2.5 inches. What is the third quartile of height in this distribution? Round your answer to two decimal places.

✓	+	−

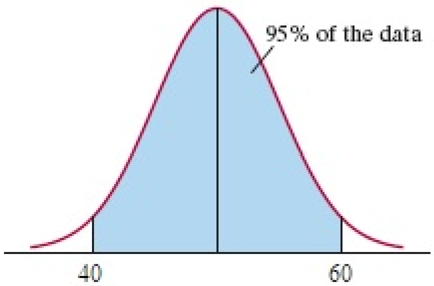
   (a) 63 inches
   (b) 65.5 inches
   (c) 69.69 inches
   (d) 72.42 inches
   (e) None of the above.

6. (5 points) Which of the following scenarios would it not be appropriate to model with a normal distribution?

✓	+	−

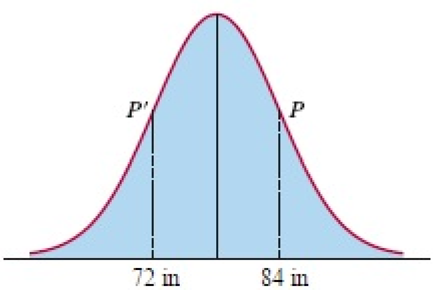
   (a) Height of an 18-year old male student at your university.
   (b) Actual weight of the contents of a 12-ounce can of dog food.
   (c) Number of fingers on the left hand of an 18-year female student at your university.
   (d) Verbal SAT score of a recent high school graduate.
   (e) None of the above.

7. (10 points) Consider the normal distribution shown. Find the mean, μ, the median, M, and the standard deviation, σ, of the distribution.



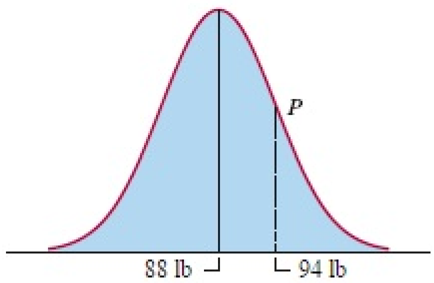
$\boxed{\checkmark \; + \; -}$

8. (10 points) Consider the normal distribution shown, and assume that P and P' are the two points of inflection of the curve. Find the median, M, the standard deviation, σ, and the third quartile, Q_3, of the distribution. $\boxed{\text{work}}$



$\boxed{\checkmark \; + \; -}$

9. (10 points) Consider the normal distribution shown and assume that P is a point of inflection of the curve. Find the mean, μ, and standard deviation, σ, of the distribution. $\boxed{\text{work}}$



✓ + −

10. (10 points) Consider a normal distribution with mean $\mu = 79$lb and standard deviation $\sigma = 88$ lb. Find the first and third quartiles of the distribution, Q_1 and Q_3. $\boxed{\text{work}}$

✓ + −

11. (10 points) Consider a normal distribution with first quartile $Q_1 = 72.625$ points and third quartile $Q_3 = 79.375$ points. Find the mean, μ, and the standard deviation, σ, of the distribution. $\boxed{\text{work}}$

$\boxed{\checkmark \quad + \quad -}$

12. (10 points) Explain why a distribution with median $M = 82$, mean $\mu = 71$, and standard deviation $\sigma = 11$ cannot be a normal distribution.

$\boxed{\checkmark \quad + \quad -}$

13. (10 points) Explain why a distribution with median $M = 426$, mean $\mu = 426$, first quartile $Q_1 = 316$, and third quartile $Q_3 = 526$ cannot be a normal distribution.

$\boxed{\checkmark \quad + \quad -}$

14. (10 points) In a normal distribution with a mean of 18, the number 25 has a standardized value of 1.6 (i.e. $x = 25$ has a z-value of 1.6). What is the standard deviation of this normal distribution? $\boxed{\text{work}}$

$\boxed{\checkmark \quad + \quad -}$

15. (25 points) After finishing college, you start working as a quality control agent at a local manufacturing plant. The machine that you are in charge of produces steel bolts. Your job is to fine tune the machine so that it produces a specific size bolt, but a small margin of error is allowable. Suppose that you believe that the machine has been calibrated so that the actual size of a bolt produced has an approximate normal distribution with a mean of 25 cm and a standard deviation of 0.001 cm. After producing 1000 bolts, you randomly choose one and find that it measures 25.008 cm. Based off of this measurement, do you think that the machine has been properly calibrated? Explain. $\boxed{\text{work}}$

$\boxed{\checkmark \quad + \quad -}$

16. (10 points (bonus)) An honest coin is tossed a total of n = 1600 times and the number of heads obtained is recorded. What is the probability that the coin lands heads greater than 840 times? Give your answer as a percentage. $\boxed{\text{work}}$

$\boxed{\checkmark \ + \ -}$

17. (10 points (bonus)) Find a MST of the following network:

$\boxed{\checkmark \ + \ -}$